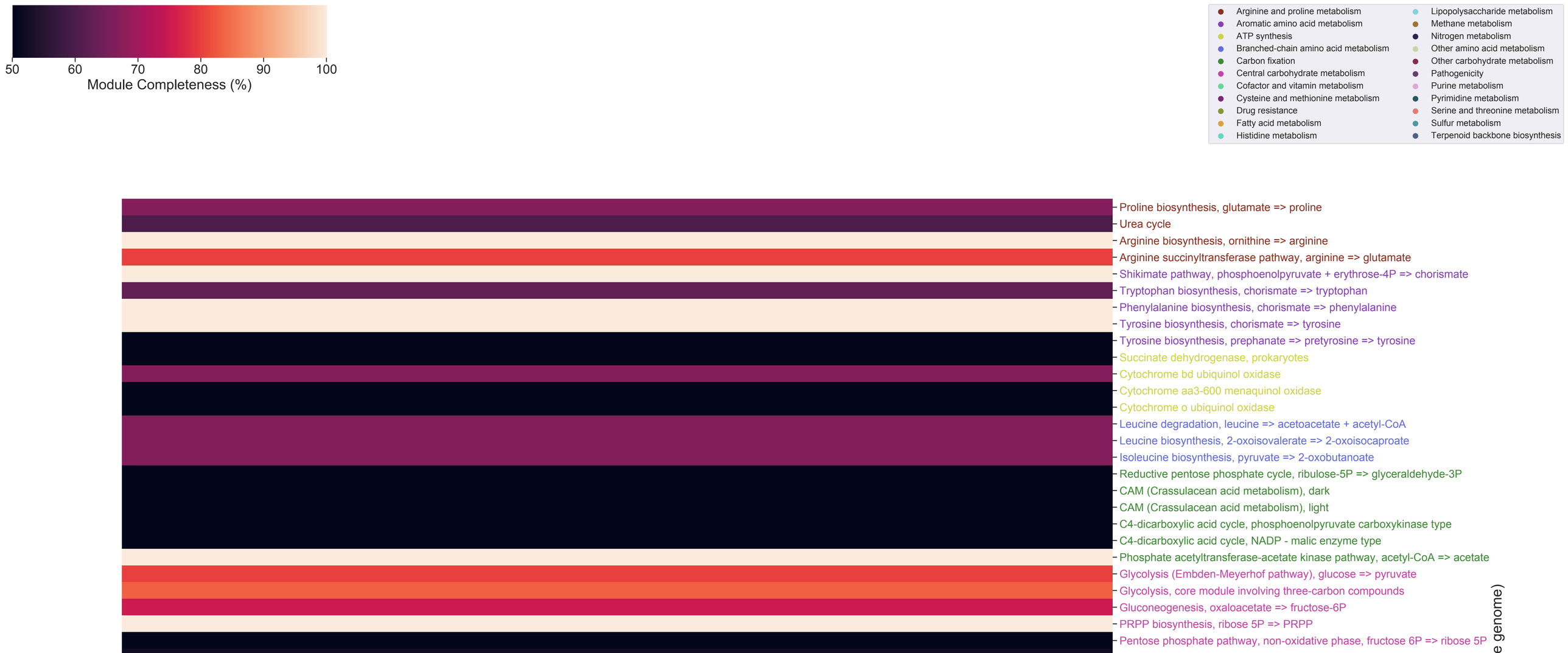
50
60
70
80
90
100
Module Completeness (%)


Arginine and proline metabolism
Aromatic amino acid metabolism
ATP synthesis
Branched-chain amino acid metabolism
Carbon fixation
Central carbohydrate metabolism
Cofactor and vitamin metabolism
Cysteine and methionine metabolism
Drug resistance
Fatty acid metabolism
Histidine metabolism
Lipopolysaccharide metabolism
Methane metabolism
Nitrogen metabolism
Other amino acid metabolism
Other carbohydrate metabolism
Pathogenicity
Purine metabolism
Pyrimidine metabolism
Serine and threonine metabolism
Sulfur metabolism
Terpenoid backbone biosynthesis
Proline biosynthesis, glutamate => proline
Urea cycle
Arginine biosynthesis, ornithine => arginine
Arginine succinyltransferase pathway, arginine => glutamate
Shikimate pathway, phosphoenolpyruvate + erythrose-4P => chorismate
Tryptophan biosynthesis, chorismate => tryptophan
Phenylalanine biosynthesis, chorismate => phenylalanine
Tyrosine biosynthesis, chorismate => tyrosine
Tyrosine biosynthesis, prephanate => pretyrosine => tyrosine
Succinate dehydrogenase, prokaryotes
Cytochrome bd ubiquinol oxidase
Cytochrome aa3-600 menaquinol oxidase
Cytochrome o ubiquinol oxidase
Leucine degradation, leucine => acetoacetate + acetyl-CoA
Leucine biosynthesis, 2-oxoisovalerate => 2-oxoisocaproate
Isoleucine biosynthesis, pyruvate => 2-oxobutanoate
Reductive pentose phosphate cycle, ribulose-5P => glyceraldehyde-3P
CAM (Crassulacean acid metabolism), dark
CAM (Crassulacean acid metabolism), light
C4-dicarboxylic acid cycle, phosphoenolpyruvate carboxykinase type
C4-dicarboxylic acid cycle, NADP - malic enzyme type
Phosphate acetyltransferase-acetate kinase pathway, acetyl-CoA => acetate
Glycolysis (Embden-Meyerhof pathway), glucose => pyruvate
Glycolysis, core module involving three-carbon compounds
Gluconeogenesis, oxaloacetate => fructose-6P
PRPP biosynthesis, ribose 5P => PRPP
Pentose phosphate pathway, non-oxidative phase, fructose 6P => ribose 5P
Citrate cycle (TCA cycle, Krebs cycle)
Citrate cycle, first carbon oxidation, oxaloacetate => 2-oxoglutarate
Pyruvate oxidation, pyruvate => acetyl-CoA
NAD biosynthesis, aspartate => NAD
Ubiquinone biosynthesis, prokaryotes, chorismate => ubiquinone
Pantothenate biosynthesis, valine--L-aspartate => pantothenate
Coenzyme A biosynthesis, pantothenate => CoA
Biotin biosynthesis, pimeloyl-ACP--CoA => biotin
Pyridoxal biosynthesis, erythrose-4P => pyridoxal-5P
Tetrahydrofolate biosynthesis, GTP => THF
Thiamine biosynthesis, AIR => thiamine-P--thiamine-2P
C1-unit interconversion, prokaryotes
C1-unit interconversion, eukaryotes
Biotin biosynthesis, BioW pathway, pimelate => pimeloyl-CoA => biotin
Molybdenum cofactor biosynthesis, GTP => molybdenum cofactor
Methionine biosynthesis, apartate => homoserine => methionine
Carbapenem resistance
Fatty acid biosynthesis, initiation
Fatty acid elongation in mitochondria
beta-Oxidation
Histidine biosynthesis, PRPP => histidine
Histidine degradation, histidine => N-formiminoglutamate => glutamate
CMP-KDO biosynthesis
ADP-L-glycero-D-manno-heptose biosynthesis
Formaldehyde assimilation, xylulose monophosphate pathway
Assimilatory nitrate reduction, nitrate => ammonia
GABA (gamma-Aminobutyrate) shunt
Glyoxylate cycle
Trehalose biosynthesis, D-glucose 1P => trehalose
Glycogen biosynthesis, glucose-1P => glycogen--starch
Glycogen degradation, glycogen => glucose-6P
ETEC pathogenicity signature, heat-labile and heat-stable enterotoxins
Inosine monophosphate biosynthesis, PRPP + glutamine => IMP
Adenine ribonucleotide biosynthesis, IMP => ADP,ATP
Guanine ribonucleotide biosynthesis IMP => GDP,GTP
Pyrimidine ribonucleotide biosynthesis, UMP => UDP--UTP,CDP--CTP
Serine biosynthesis, glycerate-3P => serine
Betaine biosynthesis, choline => betaine
Assimilatory sulfate reduction, sulfate => H2S
C5 isoprenoid biosynthesis, non-mevalonate pathway
C10-C20 isoprenoid biosynthesis, bacteria
Modules (at least 50% complete in at least one genome)
annotation.faa.ko
Genomes